STATUS UPDATE

RNSR DocDB

A relational document environment for a language-model controller: what it buys, measured layer by layer.

1,555
tests passing (unit + integration)

187
commits since Jan 2026; ~24.6k lines of Python

4
benchmark families run under one protocol: generated matters, FinanceBench, CUAD-long, OOLONG

One-line summary: **within a context window every architecture ties on accuracy**. DocDB's measurable advantages are (1) it reads documents the others cannot (scans), (2) it answers aggregation questions beyond the context window where prompting collapses, and (3) it is the only arm that produces a verified citation on every answer and a replayable derivation on computed ones. It costs 2–3× a plain REPL for that. Those three claims are now measured; the trust-tier layer still is not a signal.

WHAT WE BUILT

The system in one picture

Ingest → corpus artifact

- Docling layout parse, OCR/transcription of scanned pages, typed tables with character coordinates, FTS index, health record — one SQLite file per matter.
- Frozen: every answer binds to row ids and spans a human can re-open.

Controller in a sandboxed REPL

- Sonnet 5.5 root, Haiku 5.5 sub-calls; SQL, search ladder (FTS → embeddings), `semantic_annotate / semantic_classify`, source-bound `calculate` and `aggregate`.
- Budgets: 24 root iterations, 600 s, $2 per query; run spend ceiling.

Evidence layer

- Quotation verification, parent-executed certificates (the parent re-runs the selection SQL and re-reads every row), claim reviewer, trust tiers.

Evaluation harness (the part that grew most this month)

- Five arms over identical corpora: whole-document prompting, classic REPL, DCI (files + grep), DCI over raw files, DocDB; plus a closed-book arm (no documents).
- One-command matter bench; three public benchmarks; repeats; Wilson intervals and paired McNemar tests; two-provider judge with hand adjudication file; gold overlay with cited reasons.
- Diagnostics: auditability counters (verified citation, replayable derivation), OOLONG per-line label accuracy against gold labels.

> Design stance: the paper and the code must agree. Every number in the paper is produced by a command in the repository and pinned to a commit.

FINDINGS

What the measurements say (generated matters, 5 arms, identical corpora)

Corpus	Whole matter in prompt	Classic REPL	DCI	DCI raw files	DocDB	Reading
3 small matters (286 q, fits in window)	278	280	–	–	275	Tie. Structure buys nothing when the prompt holds the corpus.
999-doc family-law file (90 fields)	78	81	–	–	80	Tie again at ~450k tokens.
Dense uniform matter, 38 q, ~600k tokens	24	37	37	–	36–38	Prompting fails aggregation (3–4 of 15); every REPL arm writes its own parser and gets 15/15.
Dense heterogeneous matter (5 invoice layouts, 34 scans)	24	37	37	**30**	**38**	Ingestion is worth 7 questions: raw files cannot read the scans and give confident wrong totals.

38 / 38
DocDB on the heterogeneous matter, five reruns, $1.63–2.03, 5.4–5.8 min (was $2.67, 7.1 min)

12–16 / 21
computed answers with a replayable derivation from frozen source rows; no other arm produces any

2.3–2.9×
DocDB cost relative to the cheapest REPL arm on that matter; ~3× the wall time

Sources: docs/benchmarks/matters-2026-10.md; runs/dense-hetero-arms/report*.md. Judge range across passes shown in the records; all differences between REPL arms and DocDB within noise.

FINDINGS

Public benchmarks: full FinanceBench (113 q), OOLONG, CUAD-long

FinanceBench, 113 questions answered by every arm	Correct	Accuracy (95% CI)	Repeats	Spend	Verified citation	Replayable (numeric)
Closed book — question only, **no documents**	59	52% (43–61)	3 runs	$0.29	0	0 / 37
Whole document in context	95	84% (76–90)	3 runs	$42.48*	0	0 / 37
Classic REPL	96	85% (77–90)	94–96	$2.74	0	0 / 37
DCI (files + grep)	95	84% (76–90)	94–95	$3.23	0	0 / 37
DocDB	92	81% (73–88)	1 run	$9.46	**112 / 113**	**22 / 37**

- **Half of FinanceBench is answerable from memory.** The closed-book arm scores 52% (53% on numeric questions). Every reading arm is 30 points above it ($p < 0.01$, paired); no two reading arms differ significantly (largest gap classic vs DocDB +3.5 points, $p = 0.22$).
- *Whole-document prompting costs $42 on first run; repeats hit the provider's prompt cache and cost $2. The first-run figure is the honest one.
- OOLONG (350 q, synthetic classify-and-count): prompting 64% (3 runs, 221–224/350); classic REPL 81% on the 182 it completed. DocDB's earlier 50-question run: 36/50 at $2.90 after the label-once change (was 34/50 at $16.77).
- CUAD-long (10 contracts): 7 / 6 / 6 — nothing distinguishable on n = 10.

Provisional. These FinanceBench/OOLONG rows are judged by Gemini 3.8 Flash alone (3 passes) because the Anthropic account reached its monthly API limit on 23 Oct; the Haiku judge, the DocDB repeats and the remaining OOLONG arms resume on 1 Nov. Judge version 4 ("a matching verdict word with a different value is NO") is being applied now; the numbers above are v3 and may move by a few questions.

PROGRESS

What changed in October (and what each change was worth)

Change	Why	Measured effect
Aggregate certificate (`aggregate` over parent-executed SQL, uncapped operands)	Reviewer withheld trust from correct sums; 32-operand cap was arbitrary	Replayable derivations 8–15 → 12–16 of 21 computed questions
Scope-only reviewer for certificates; embedding gate on the search ladder	Reviewer argued with arithmetic it could not check; 10,554 embedding requests bought nothing	Requests 10,554 → 223 on the heterogeneous matter; 38/38
Review loop capped at 3 rounds; certificate skip; certificate nudge	Six full-context root calls per contested field	Root calls –33%, spend $2.67 → $1.63–2.03, no accuracy change
Label once, certify the count (OOLONG classification)	Three votes per label, then voting with its own reviewer	$16.77 → $2.90, 6,915 → 895 sub-calls, 11 → 0 halts, 34 → 36/50
Answer presentation: number first, bounded precision	Certificate path carried its proof in the answer text	FinanceBench subset 30 → 35/40 against the same July answers
Judge v3/v4, normaliser, two providers, adjudication file	Bare-token judge penalised correct numbers stated with a formula; one judge family is not an instrument	Every arm moved 5–12 questions between judge versions; Gemini accepts self-corrected answers Haiku rejected — now a stated policy
Count certificates over text columns; bounded proofs for the reviewer	Controller refused 4× trying to certify a count; 317k-char proofs were dropped unread	Largest proof 317k → 36k chars; fields now judged instead of dropped

STATUS BY WORKSTREAM

Where each piece stands

Workstream	State	Notes
Ingestion (Docling, OCR, typed tables, health)	Done	Measured value: documents a model cannot otherwise read. Zero value on clean 10-Ks.
Sandboxed controller + tools	Done	Parity with every REPL baseline on every corpus; the only arm with a 15/15 and a 38/38.
Aggregate certificate / source-bound calculation	Done	Works where tables exist. Gap: 13% of invoices on the hetero matter have no table → certificate cannot reach them.
Claim review loop	Bounded, not solved	19–23 of 38 correct answers still finish low-tier: reviewer cannot verify scope of controller-chosen operands.
Trust tiers	Not yet a signal	High tier reached on 2 of 113 FinanceBench answers (first time); needs table validation at ingest.
Evaluation instrument (repeats, CIs, two judges, overlay, strata)	Built	Code complete and tested; first full runs partly complete (see blocker).
Full public-benchmark runs (6 arms × 3 repeats)	Blocked until 1 Nov	FinanceBench: 4 arms × 3 + DocDB × 1 done. OOLONG: prompting × 3 done, classic 182/350, DocDB 0/350. Resume is automatic (same commands).
Part B score work (verdict-first rubric; specific-before-generic labels)	On branch	Diagnosed from label data: 93% label accuracy, "entity" absorbing places/people/concepts → exact counts fail. Fix written on `partb-scores`, unmeasured.
Paper (arXiv draft, HTML)	v2 drafted	§6.7 and Table 6 need the full-run numbers; pinned to commits 3d62d4b / 1b990a8.

ASSESSMENT

Is it valuable? What the evidence supports, and what it does not

Supported by measurement

- Beyond the context window, a REPL (any REPL) beats prompting on aggregation 15/15 vs 3–4/15. That part is commodity: DCI and classic get it too.
- Layout-aware ingestion is worth ~18% of questions on a corpus with scans, and 0% on clean PDFs.
- DocDB is the only arm with a verified citation on every answer and a replayable derivation on 54–76% of computed ones. This is the product property; no baseline has it.
- Cost of that property: 2–3× a REPL and ~3× its wall time, down from ~10× in September.

Not supported (yet)

- No accuracy advantage over a plain REPL anywhere. Claims of "more accurate" should not be made.
- Trust tiers do not predict correctness: on the dense matter 22/38 correct answers are low-tier because the reviewer cannot check scope.
- Half of FinanceBench is in the model's memory; the benchmark measures recall as much as reading. Any public-10-K result needs the closed-book baseline beside it.
- Judge choice moves scores by 5–12 questions on 40. Differences of a few questions are not findings.

> So what: the pitch is **auditability at a known price**, not accuracy. The relational layer is what makes a derivation replayable; the question for legal-tech is whether a receipt a reader can re-open is worth 2–3× the inference cost. That is now a priced, product question rather than a research one.

BLOCKERS AND RISKS

What is in the way

Blocker

- **Anthropic monthly API limit reached 23 Oct** ("regain access 2026-11-01"). All Claude-based runs and the Haiku judge are paused. Raising the limit in the console would unblock immediately; otherwise everything resumes 1 Nov with the same commands (completed rows are kept, failed rows retried).

Risks

- Remaining full-run spend ≈ $90 (FinanceBench DocDB ×2, OOLONG classic ×3, OOLONG DocDB ×3) plus ~$6 judge; Part B reruns another ≈ $70.
- Gemini judge is more lenient than Haiku on self-correcting answers; cross-judge disagreement will land in the adjudication file and needs a human reader (20-row sample ready).
- Certificates cannot cover numbers that live in prose (no table). Structural fix: text-span rows in `aggregate()`. Not started.

Spend to date (list price, October)

Small matters (3 corpora, 3 arms)	~$38
TS6 + dense matters, all reruns	~$50
Public 40/10/50 subsets, 5 arms (incl. superseded runs)	~$80
Full FinanceBench/OOLONG so far	~$93
Judging (both providers)	~$10

October total ≈ $270 at list price, summed from each run's results; per-run figures are in each run's summary.json. Haiku 5.5 is priced at its ≤100k-token tier so long-prompt sub-calls are under-counted.

NEXT

Next four weeks

1. **1 Nov — resume the full runs.** FinanceBench DocDB ×2 more, OOLONG classic and DocDB ×3; re-judge everything with both providers; mark the adjudication sample by hand; publish intervals, not points.
2. **Merge Part B and measure it.** Verdict-first answers; classification rubrics from most specific to catch-all label. Expect OOLONG exact-count questions to move; expect FinanceBench not to.
3. **Text-span rows in the aggregate certificate**, so a parent-executed selection can cover a corpus whose numbers are partly in prose; then re-measure how many correct answers still finish low-tier.
4. **Table validation at ingest**, so the high tier is reachable and tier calibration can be measured on held-out matters.
5. **A corpus of real scanned vendor paperwork.** The ingestion result rests on generated scans; the claim needs real ones.
6. **Paper to arXiv** once items 1–2 land: Table 6 and §6.7 from the full runs, closed-book baseline in every public table, judge protocol and adjudication agreement reported.

> Decision requested: raise the Anthropic monthly limit now (unblocks ~1 week early), or hold to 1 Nov.